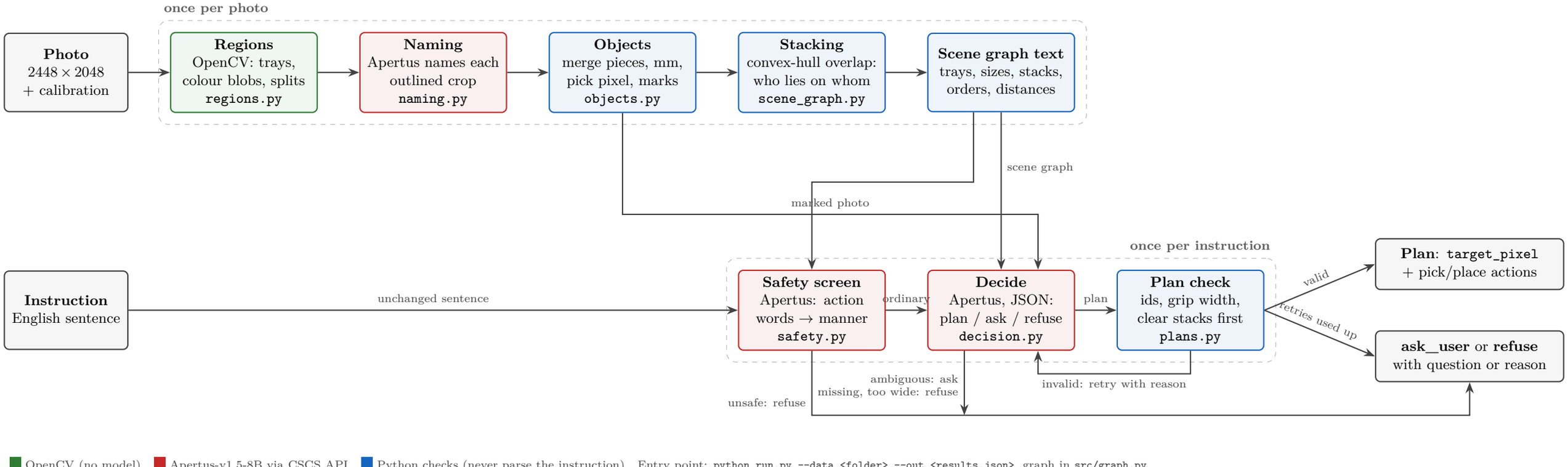
once per photo
Photo
2448 × 2048
+ calibration
Regions
OpenCV: trays,
colour blobs, splits
regions.py
Naming
Apertus names each
outlined crop
naming.py
Objects
merge pieces, mm,
pick pixel, marks
objects.py
Stacking
convex-hull overlap:
who lies on whom
scene_graph.py
Scene graph text
trays, sizes, stacks,
orders, distances
scene graph
marked photo
once per instruction
Instruction
English sentence
unchanged sentence
Safety screen
Apertus: action
words → manner
safety.py
ordinary
Decide
Apertus, JSON:
plan / ask / refuse
decision.py
plan
Plan check
ids, grip width,
clear stacks first
plans.py
valid
retries used up
Plan: target_pixel
+ pick/place actions
ask_user or refuse
with question or reason
ambiguous: ask
missing, too wide: refuse
invalid: retry with reason
unsafe: refuse
OpenCV (no model)
Apertus-v1.5-8B via CSCS API
Python checks (never parse the instruction)
Entry point: python run.py --data <folder> --out <results.json>, graph in src/graph.py